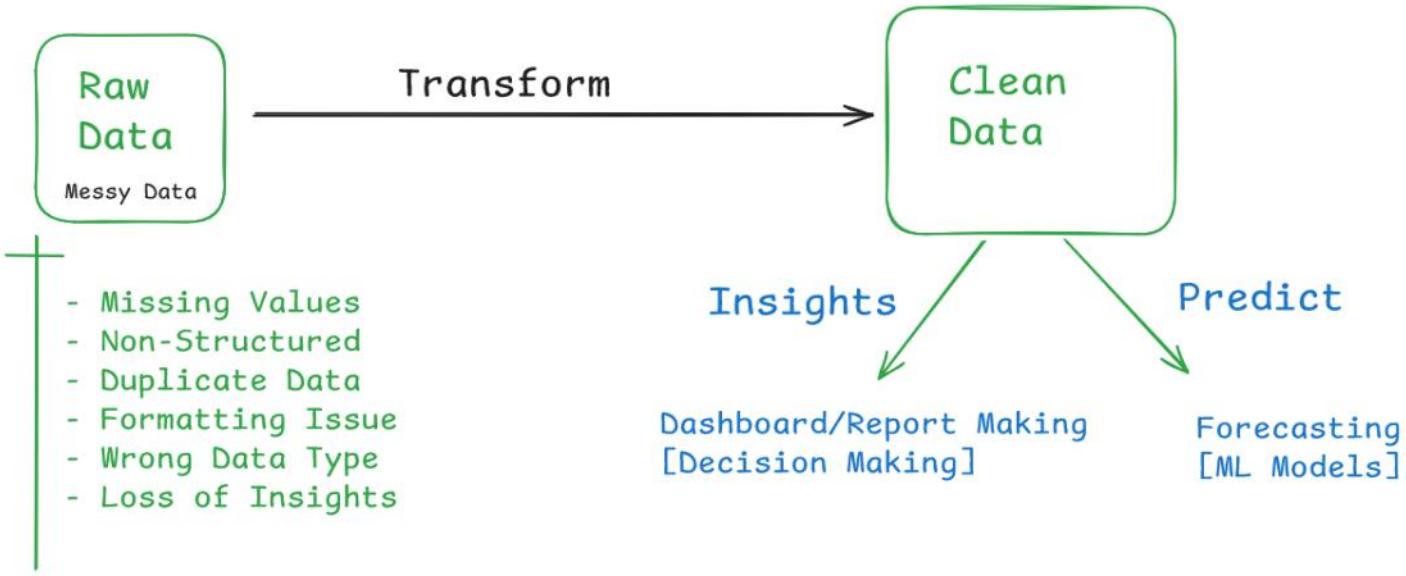
Introduction To SQL

🎯 Session Objectives

- ✅ Understand what databases are and why they matter
- ✅ Differentiate between various types of databases
- ✅ Grasp the importance and use of SQL in real-world scenarios
- ✅ Get familiar with SQL installation and the MySQL Workbench interface
- ✅ Import datasets into MySQL Workbench
- ✅ Use SELECT statements to explore data
- ✅ Understand and apply the DESCRIBE function
- ✅ Recognize SQL data types 🧠
- ✅ Use the WHERE clause for filtering 🔍

```
Binary Format - 0/1
Github - Lecture Notes
[Concept Notes]
[.sql file]
.csv[Comma Separated File]
```



What is a Data?

In Today's World, Data is the new oil to the industry.
Data is Knowledge , Data is Money.....

What is a Database...

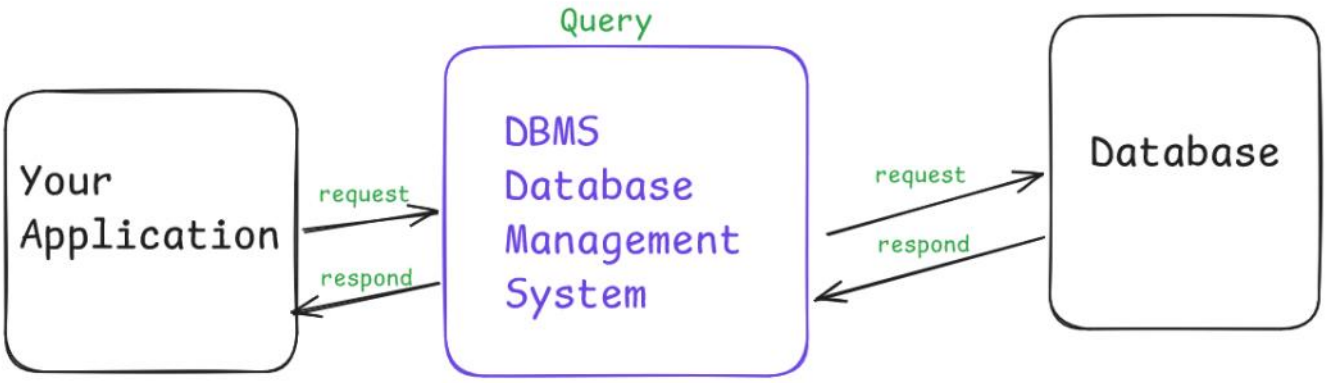
Collection of Data [Tabular Format]
[Rows * Cols]

Database

Query [SQL]
[Structured Query Language]

Person A [French]
Person B [Hindi]

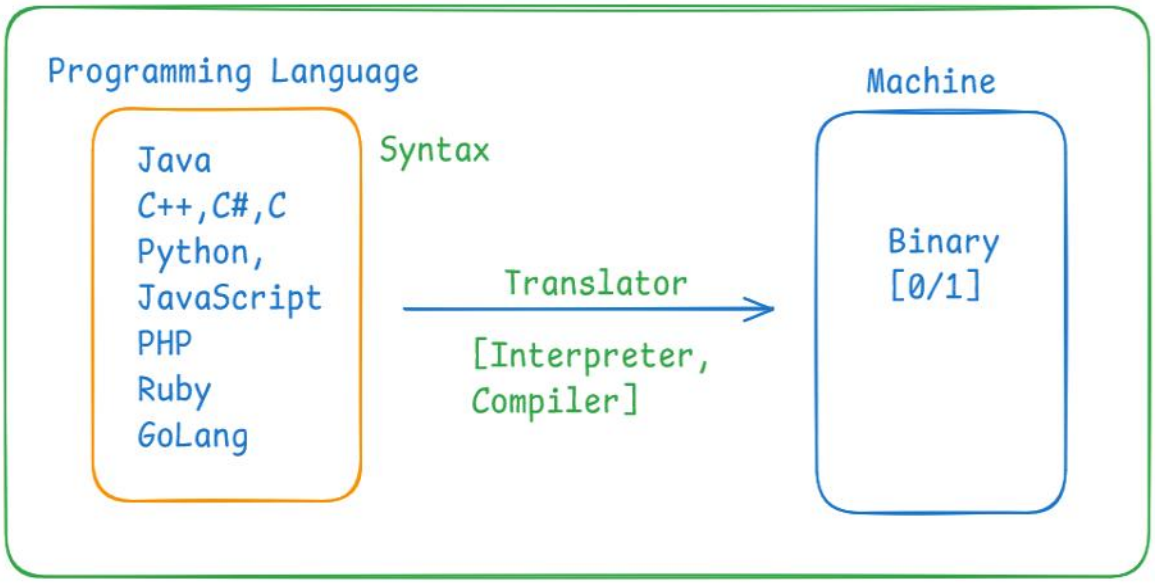
Mode of Communication between 2 or more people on common ground.

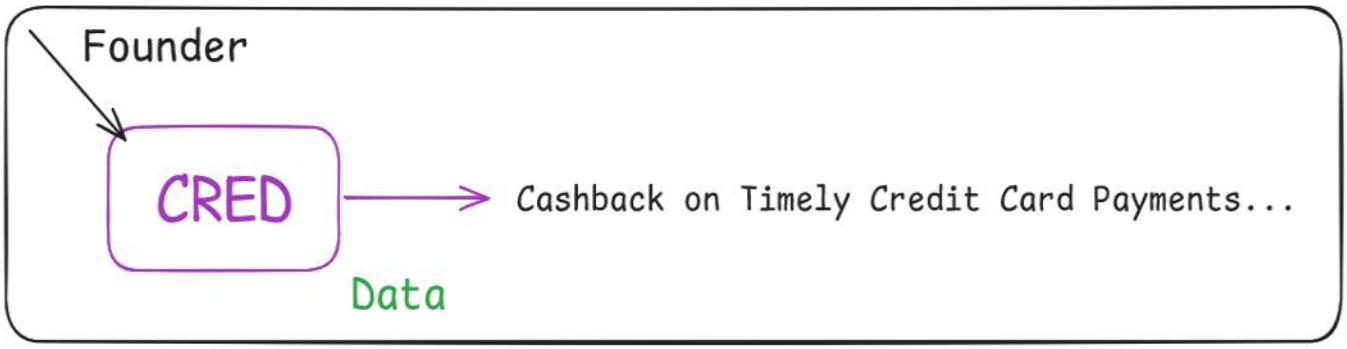


R-DBMS [Relational]

- MySQL
- PostgreSQL
- SQLite
- Oracle
- SQL Server

Structured Data...



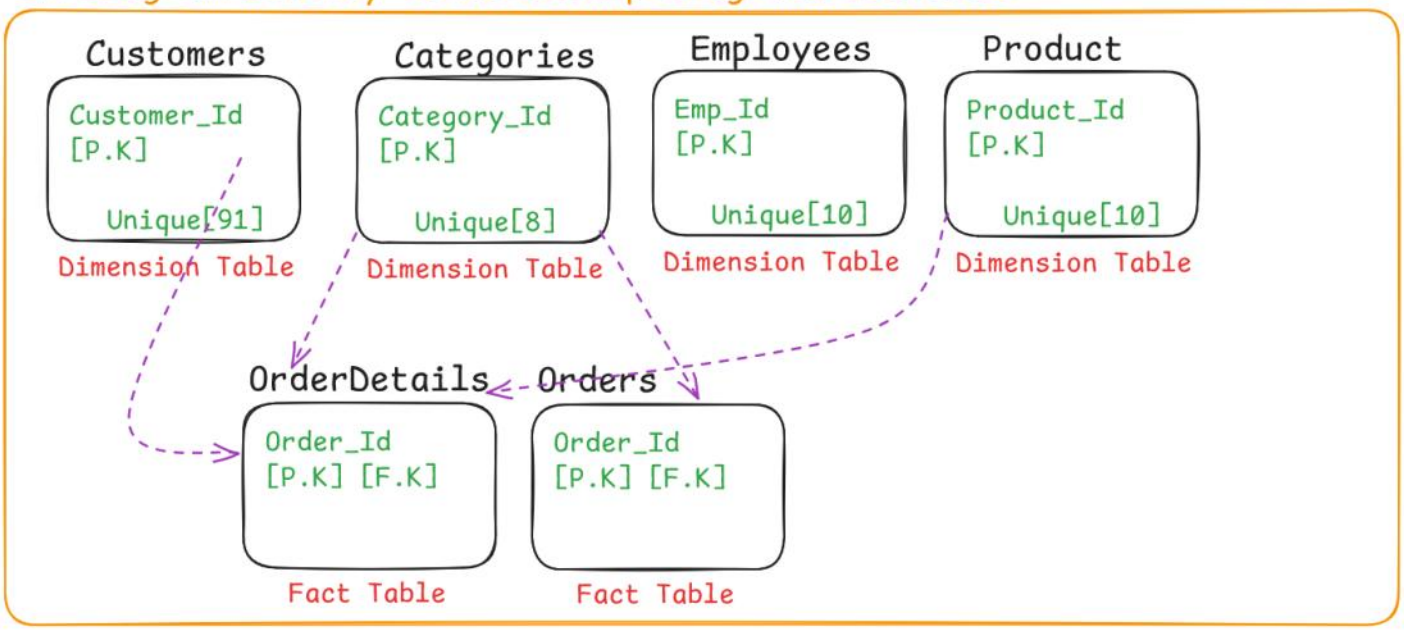


Practical Example

Your Database:

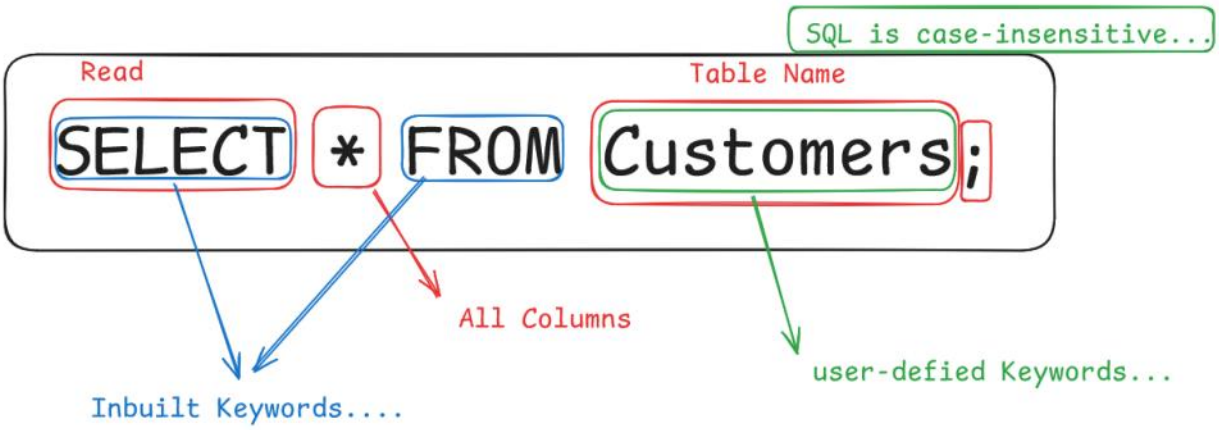
Tablenames	Records
Customers	91
Categories	8
Employees	10
OrderDetails	518
Orders	196
Products	77
Shippers	3
Suppliers	29

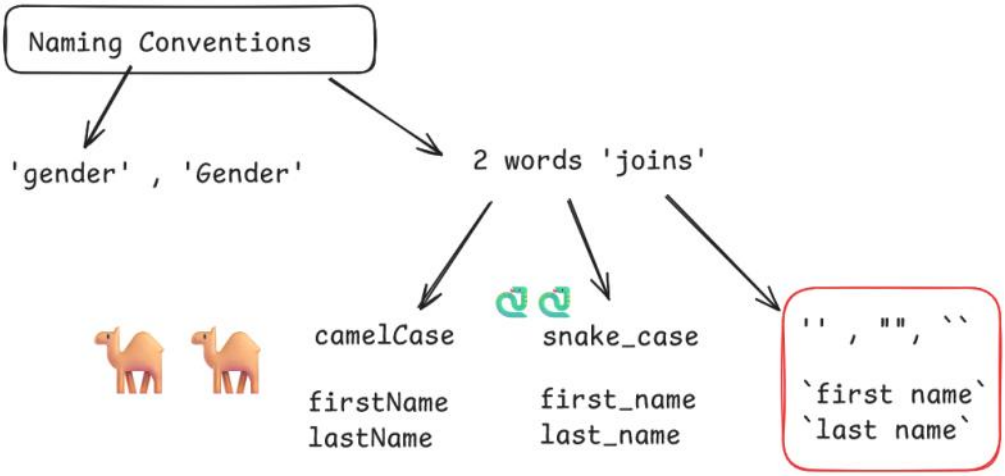
ER Diagram [Entity Relationship Diagram] [Schema]



Dimension Table - Unique Values + [Primary Keys] [No Duplicates]

Fact Table - Foreign Keys + Transactional Records





MySQL Workbench

```
SELECT
    CustomerName,
    Address,
    City
FROM Customers;
```

Column Name

Number of Records: 91

CustomerName	Address	City
Alfreds Futterkiste	Obere Str. 57	Berlin
Ana Trujillo Emparedados y helados	Avda. de la Constitución 2222	México D.F.
Antonio Moreno Taquería	Mataderos 2312	México D.F.
Around the Horn	120 Hanover Sq.	London
Berglunds snabbköp	Berguvsvägen 8	Luleå
Blauer See Delikatessen	Forsterstr. 57	Mannheim
Blondel père et fils	24, place Kléber	Strasbourg
Bólido Comidas preparadas	C/ Araquil, 67	Madrid

```
SELECT
    CustomerName,
    City,
    Country,
    Address
FROM Customers;
```

Number of Records: 91

CustomerName	City	Country	Address
Alfreds Futterkiste	Berlin	Germany	Obere Str. 57
Ana Trujillo Emparedados y helados	México D.F.	Mexico	Avda. de la Constitución 2222
Antonio Moreno Taquería	México D.F.	Mexico	Mataderos 2312
Around the Horn	London	UK	120 Hanover Sq.
Berglunds snabbköp	Luleå	Sweden	Berguvsvägen 8
Blauer See Delikatessen	Mannheim	Germany	Forsterstr. 57
Blondel père et fils	Strasbourg	France	24, place Kléber
Bólido Comidas preparadas	Madrid	Spain	C/ Araquil, 67

First 5 Product from Products Tables

LIMIT 5

Row Filter

'Where' Clause

```
SELECT
    ProductName,
    Unit,
    Price
FROM Products
WHERE Price >= 25;
```

Number of Records: 29

ProductName	Unit	Price
Grandmas Boysenberry Spread	12 - 8 oz jars	25.00
Uncle Bobs Organic Dried Pears	12 - 1 lb pkgs.	30.00
Northwoods Cranberry Sauce	12 - 12 oz jars	40.00
Mishi Kobe Niku	18 - 500 g pkgs.	97.00
Ikura	12 - 200 ml jars	31.00
Queso Manchego La Pastora	10 - 500 g pkgs.	38.00
Alice Mutton	20 - 1 kg tins	39.00
Carnarvon Tigers	16 kg pkg.	62.50

Royal En-field Dataset
[Bike Sales Analysis]

MySQL Workbench Installation

Mac User

Intel Chip
Apple Silicon

https://dev.mysql.com/downloads/installer/

MySQL Community Downloads

MySQL Installer

General Availability (GA) Releases | Archives

MySQL Installer 8.0.45

Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:

8.0.45

Select Operating System:

Microsoft Windows

Windows (x86, 32-bit), MSI Installer	8.0.45	2.1M	Download
(mysql-installer-web-community-8.0.45.0.msi)		MD5: c221d45c90c003b585b9570edc46a8ad | Signature	
Windows (x86, 32-bit), MSI Installer	8.0.45	556.0M	Download
(mysql-installer-community-8.0.45.0.msi)		MD5: 2e6a5e6e9425700a3d066f185992e2ad | Signature	

MySQL Community Downloads

Login Now or Sign Up for a free account.

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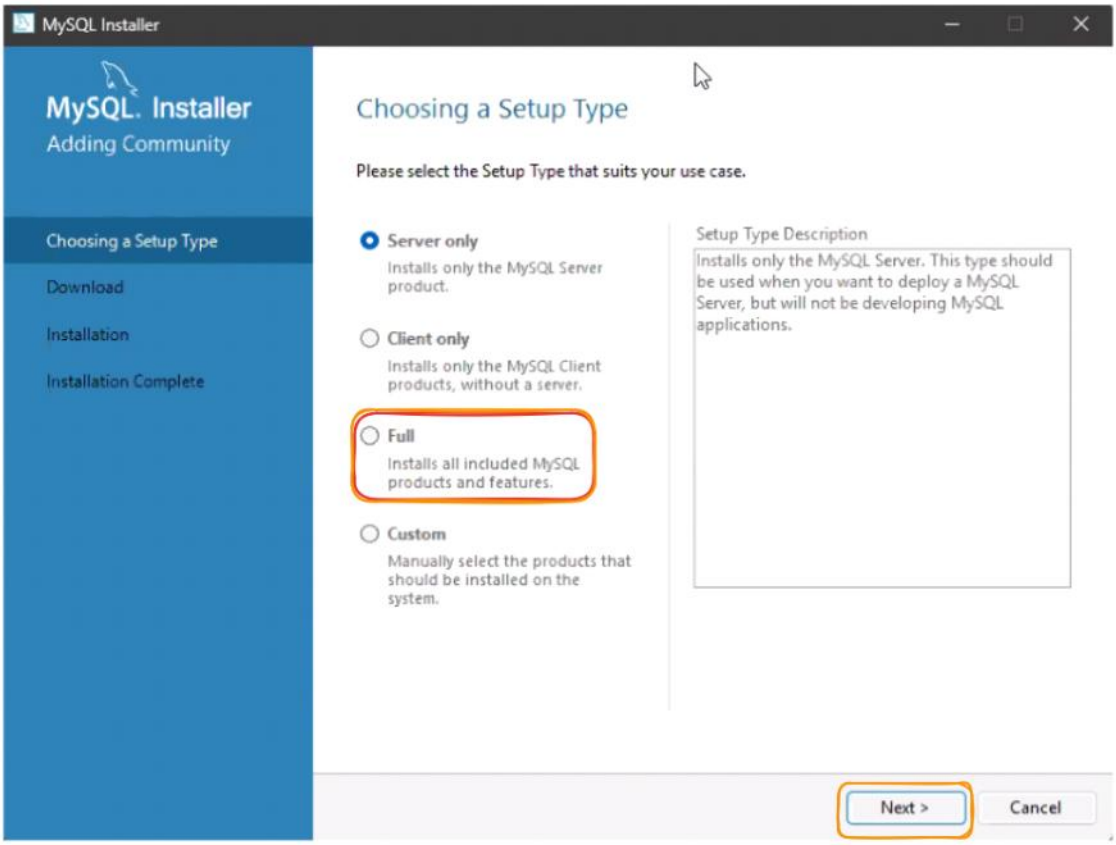
- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

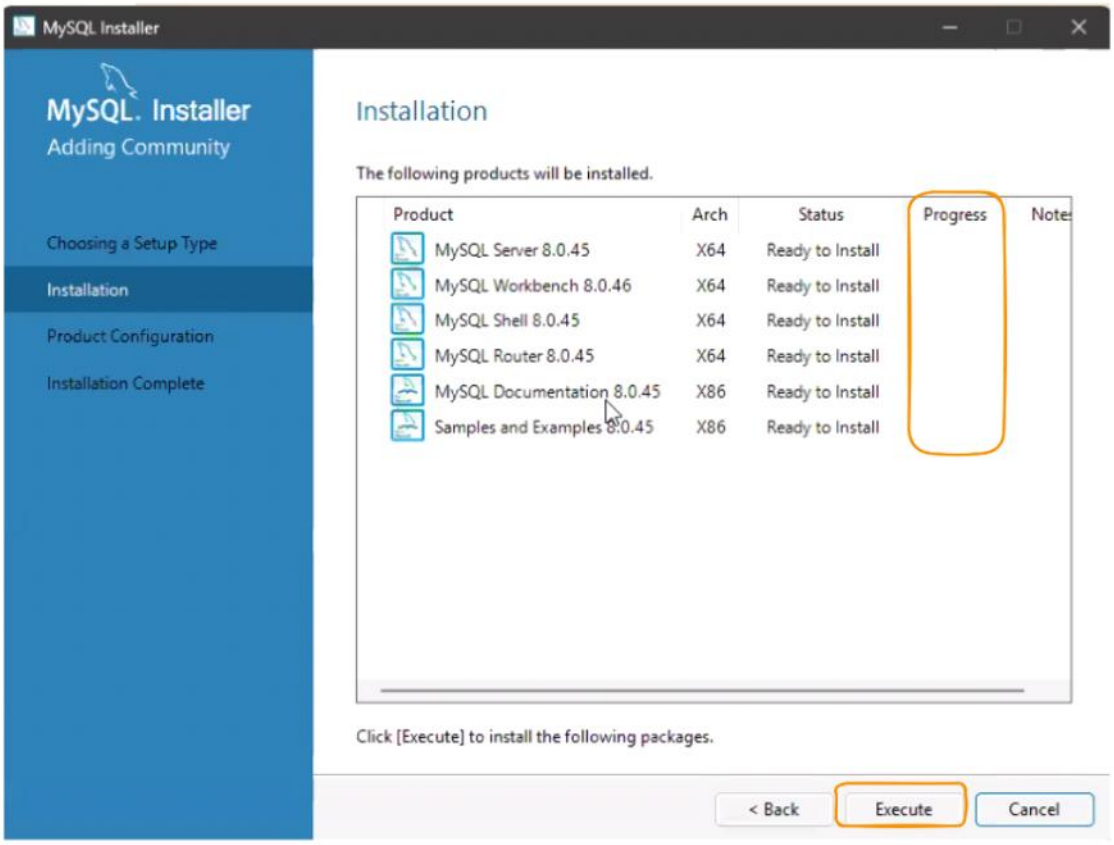
Login » using my Oracle Web account

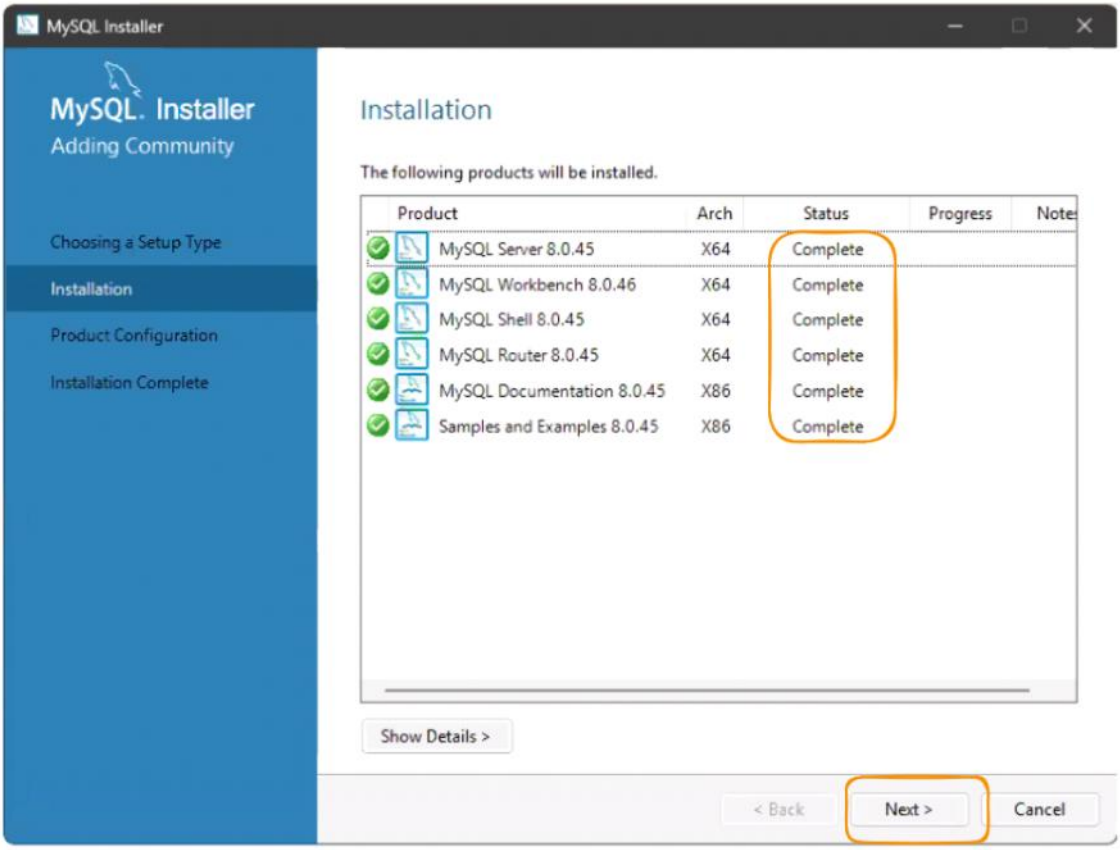
Sign Up » for an Oracle Web account

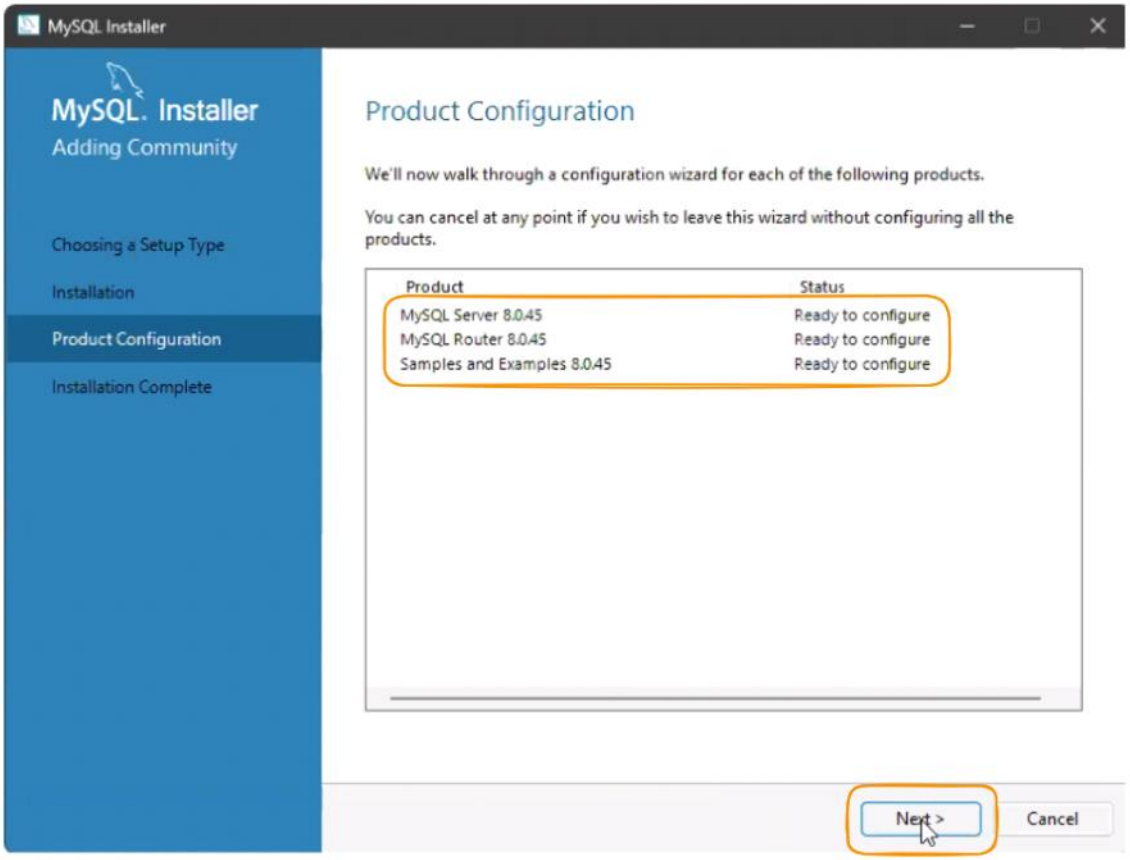
MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

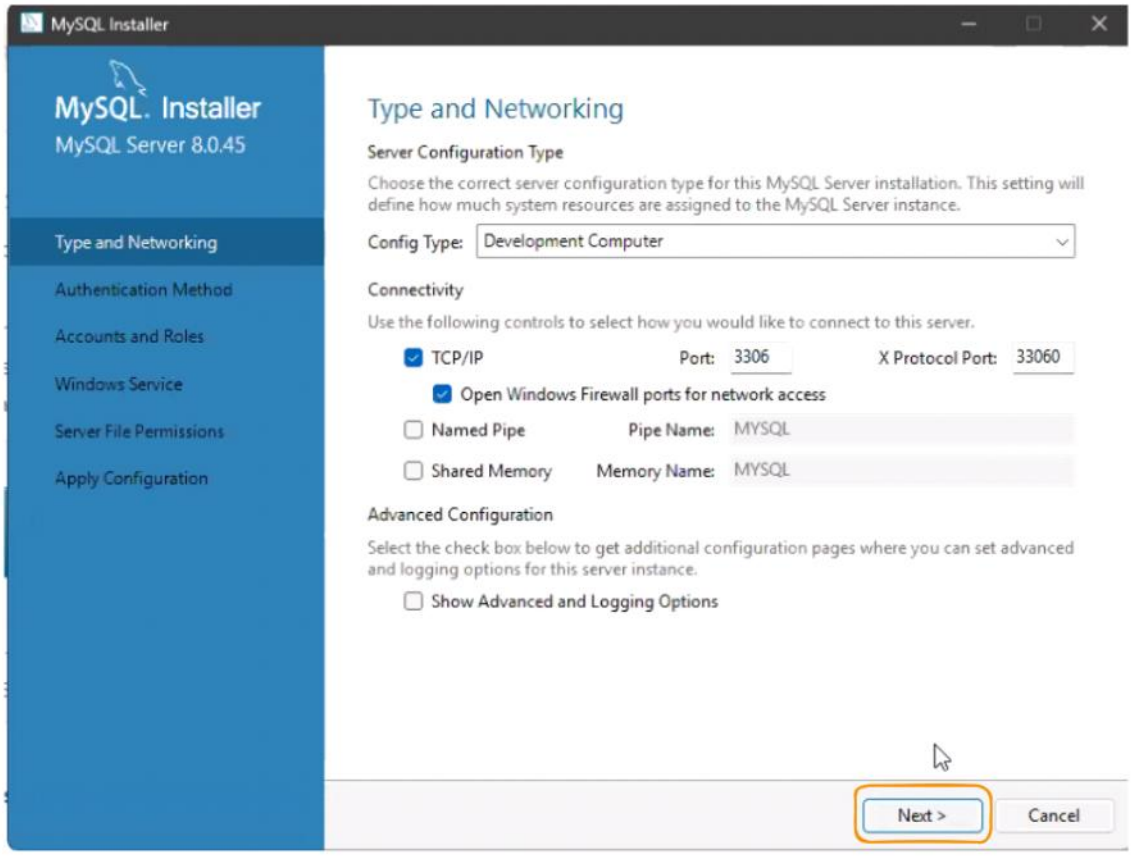
No thanks, just start my download.

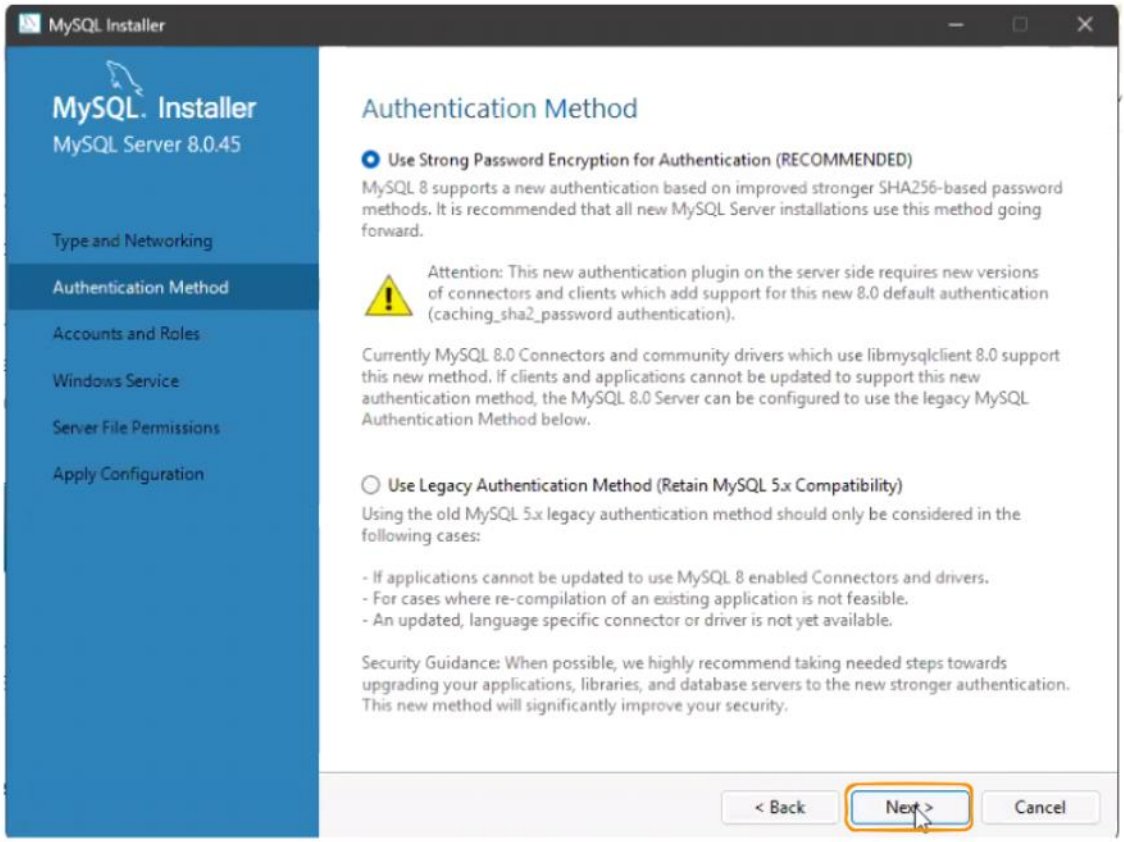

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Download
Installation
Installation Complete
Choosing a Setup Type
Please select the Setup Type that suits your use case.
Server only
Installs only the MySQL Server product.
Client only
Installs only the MySQL Client products, without a server.
Full
Installs all included MySQL products and features.
Custom
Manually select the products that should be installed on the system.
Setup Type Description
Installs only the MySQL Server. This type should be used when you want to deploy a MySQL Server, but will not be developing MySQL applications.
Next >
Cancel

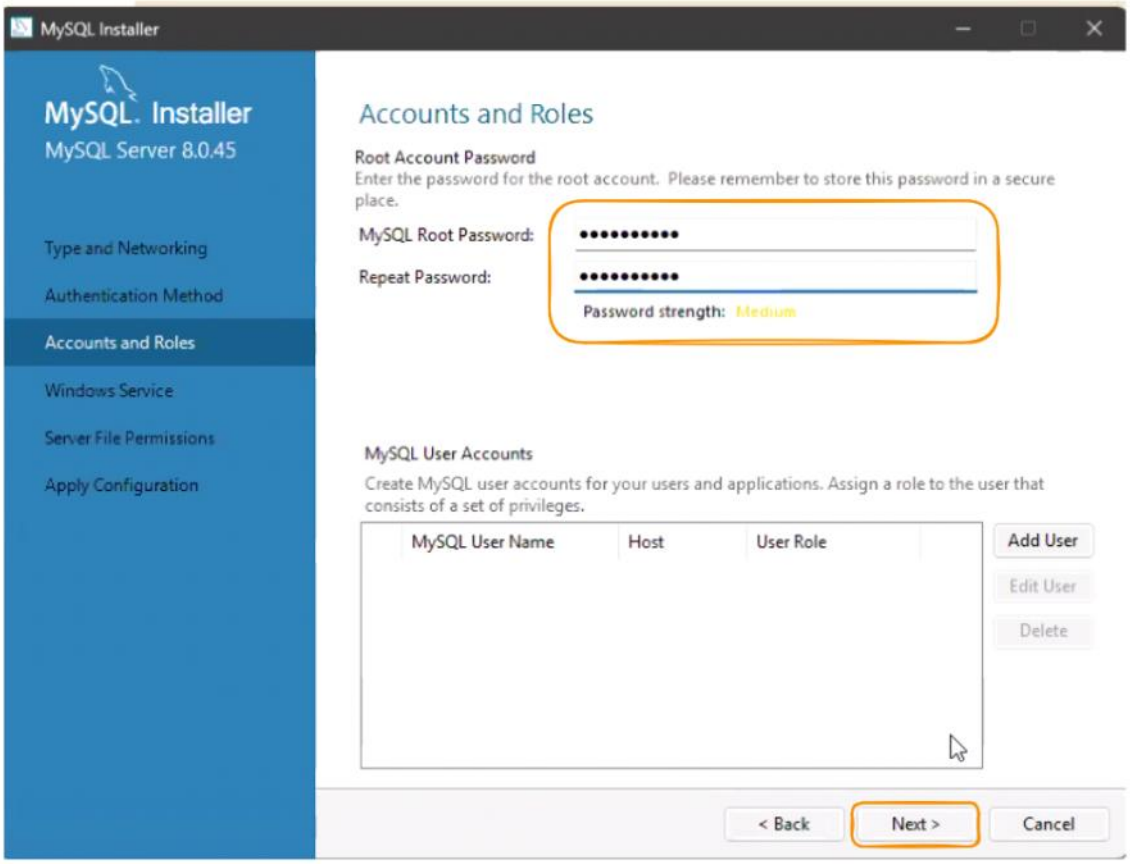

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Installation
The following products will be installed.
Product
Arch
Status
Progress
Notes
MySQL Server 8.0.45
X64
Ready to Install
MySQL Workbench 8.0.46
X64
Ready to Install
MySQL Shell 8.0.45
X64
Ready to Install
MySQL Router 8.0.45
X64
Ready to Install
MySQL Documentation 8.0.45
X86
Ready to Install
Samples and Examples 8.0.45
X86
Ready to Install
Click [Execute] to install the following packages.
< Back
Execute
Cancel

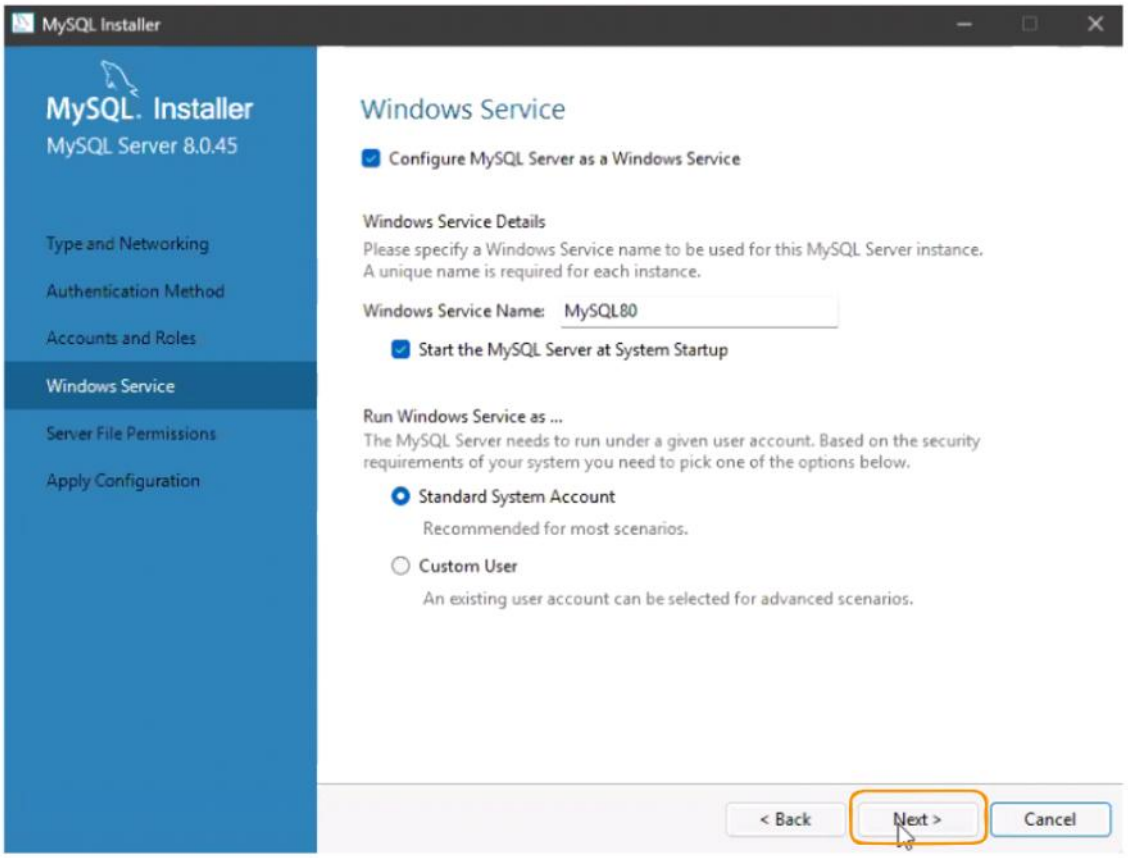

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Installation
The following products will be installed.
Product
Arch
Status
Progress
Notes
MySQL Server 8.0.45
X64
Complete
MySQL Workbench 8.0.46
X64
Complete
MySQL Shell 8.0.45
X64
Complete
MySQL Router 8.0.45
X64
Complete
MySQL Documentation 8.0.45
X86
Complete
Samples and Examples 8.0.45
X86
Complete
Show Details >
< Back
Next >
Cancel

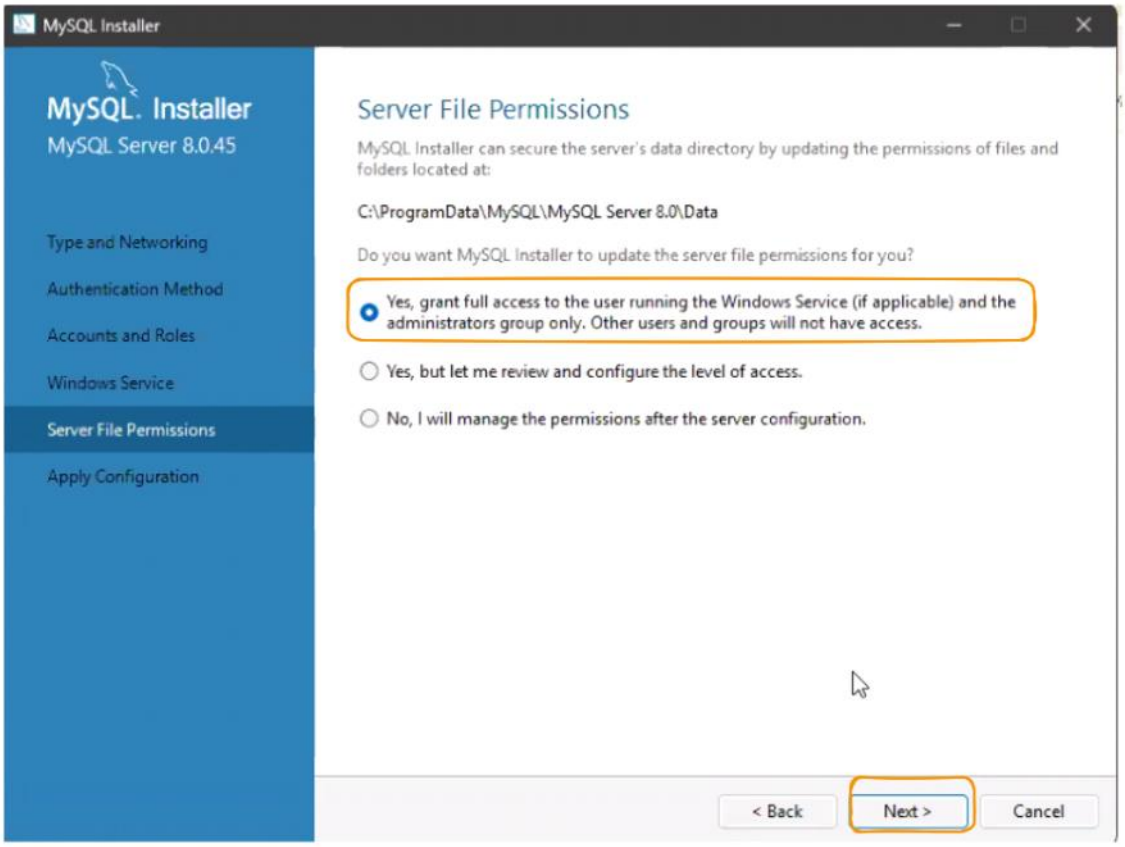

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.45
Ready to configure
MySQL Router 8.0.45
Ready to configure
Samples and Examples 8.0.45
Ready to configure
Next >
Cancel

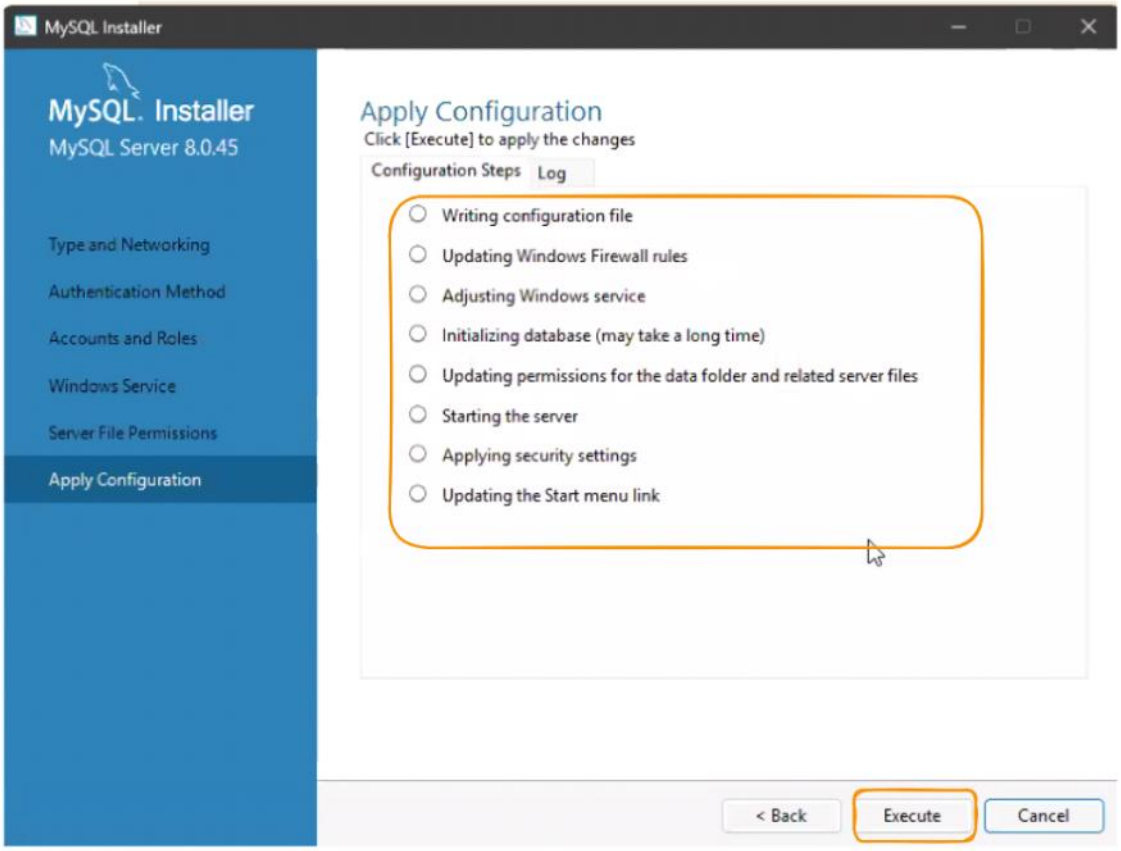

MySQL Installer
MySQL. Installer
MySQL Server 8.0.45
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Type and Networking
Server Configuration Type
Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.
Config Type: Development Computer
Connectivity
Use the following controls to select how you would like to connect to this server.
TCP/IP
Port: 3306
X Protocol Port: 33060
Open Windows Firewall ports for network access
Named Pipe
Pipe Name: MYSQL
Shared Memory
Memory Name: MYSQL
Advanced Configuration
Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.
Show Advanced and Logging Options
Next >
Cancel

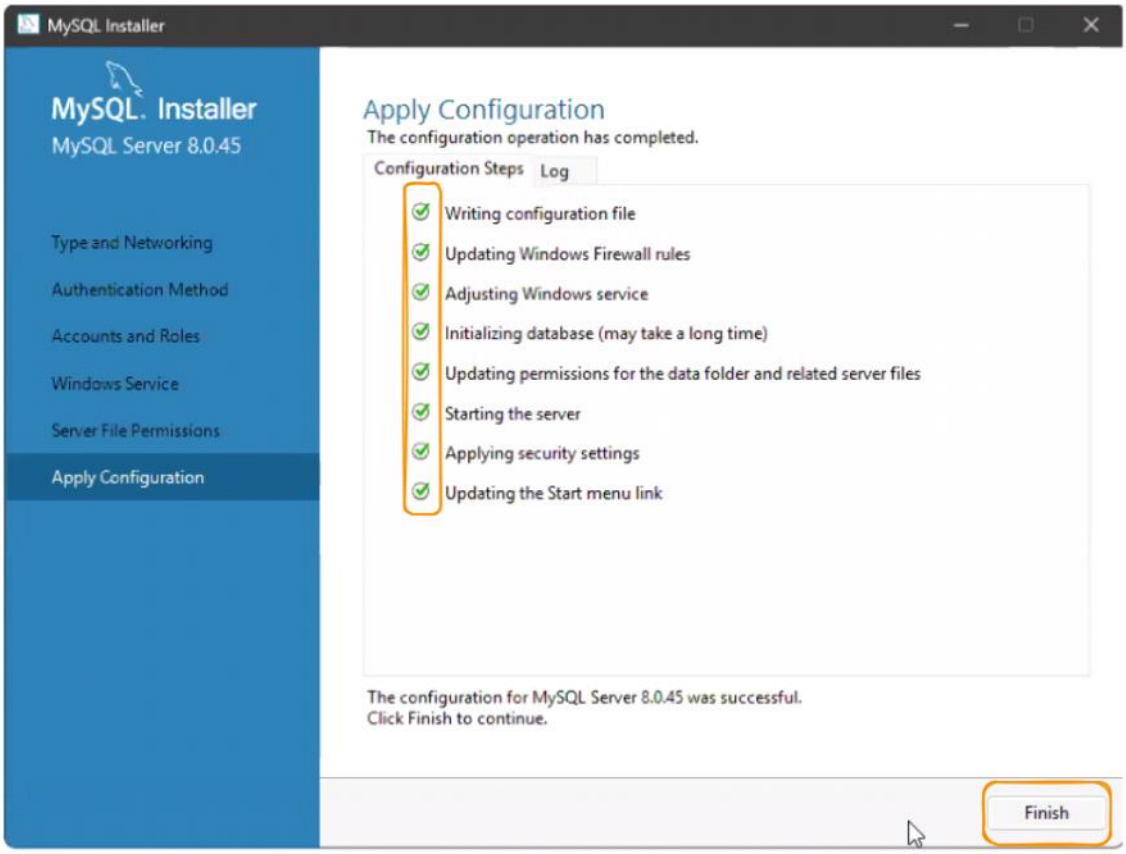

MySQL Installer
MySQL. Installer
MySQL Server 8.0.45
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Authentication Method
Use Strong Password Encryption for Authentication (RECOMMENDED)
MySQL 8 supports a new authentication based on improved stronger SHA256-based password methods. It is recommended that all new MySQL Server installations use this method going forward.
Attention: This new authentication plugin on the server side requires new versions of connectors and clients which add support for this new 8.0 default authentication (caching_sha2_password authentication).
Currently MySQL 8.0 Connectors and community drivers which use libmysqlclient 8.0 support this new method. If clients and applications cannot be updated to support this new authentication method, the MySQL 8.0 Server can be configured to use the legacy MySQL Authentication Method below.
Use Legacy Authentication Method (Retain MySQL 5.x Compatibility)
Using the old MySQL 5.x legacy authentication method should only be considered in the following cases:
- If applications cannot be updated to use MySQL 8 enabled Connectors and drivers.
- For cases where re-compilation of an existing application is not feasible.
- An updated, language specific connector or driver is not yet available.
Security Guidance: When possible, we highly recommend taking needed steps towards upgrading your applications, libraries, and database servers to the new stronger authentication. This new method will significantly improve your security.
< Back
Next >
Cancel

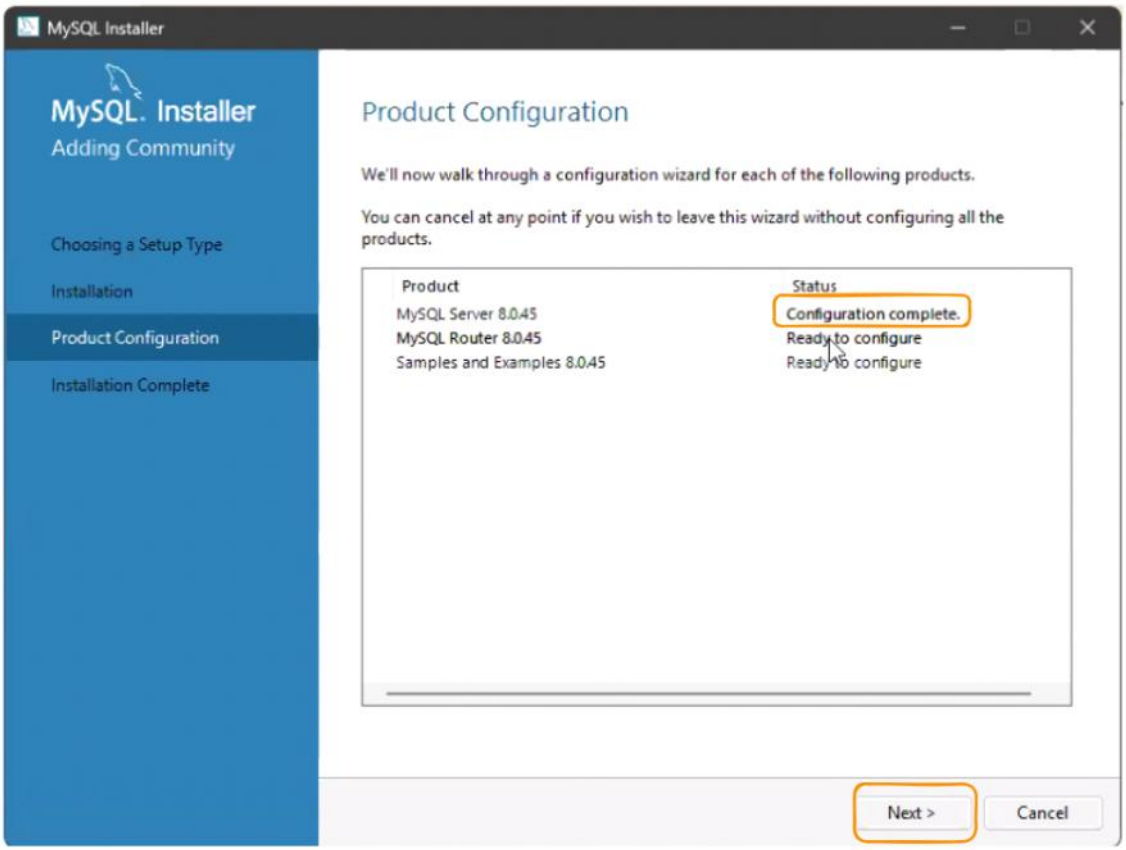

MySQL Installer
MySQL. Installer
MySQL Server 8.0.45
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Accounts and Roles
Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.
MySQL Root Password:
Repeat Password:
Password strength: Medium
MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.
MySQL User Name
Host
User Role
Add User
Edit User
Delete
< Back
Next >
Cancel

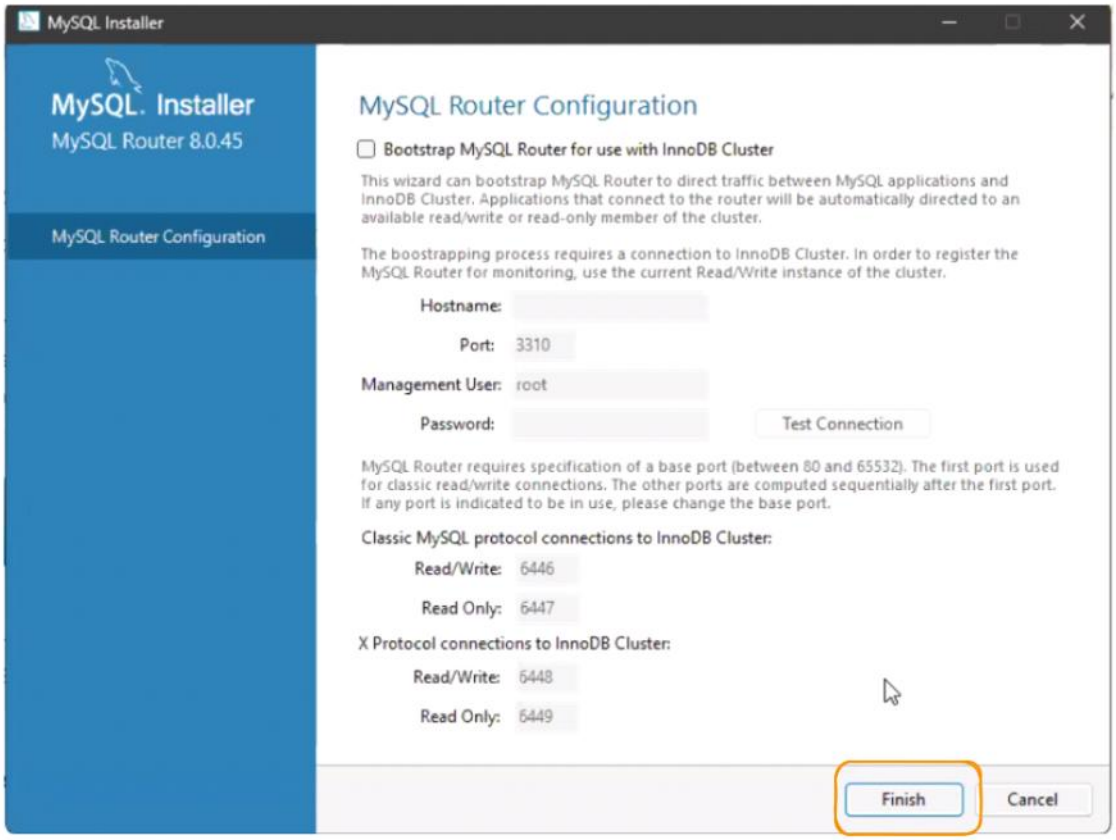

MySQL Installer
MySQL. Installer
MySQL Server 8.0.45
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Windows Service
Configure MySQL Server as a Windows Service
Windows Service Details
Please specify a Windows Service name to be used for this MySQL Server instance.
A unique name is required for each instance.
Windows Service Name: MySQL80
Start the MySQL Server at System Startup
Run Windows Service as ...
The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.
Standard System Account
Recommended for most scenarios.
Custom User
An existing user account can be selected for advanced scenarios.
< Back
Next >
Cancel

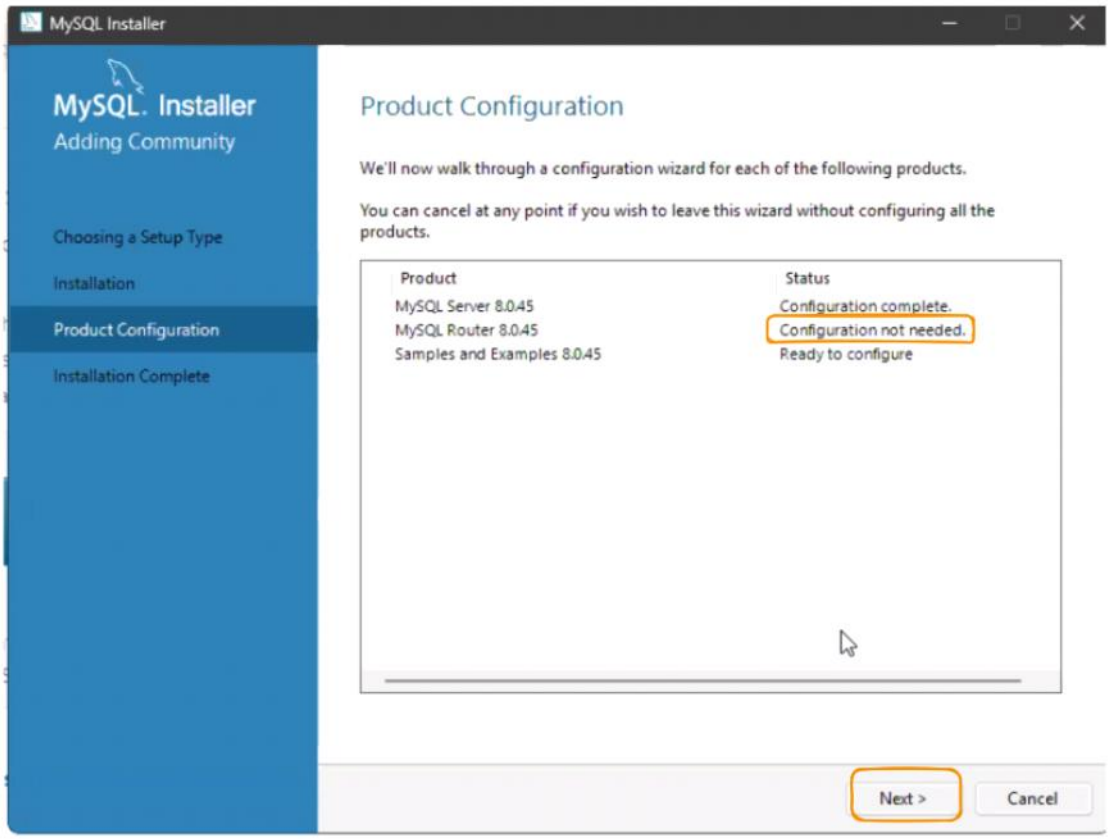

MySQL Installer
MySQL. Installer
MySQL Server 8.0.45
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Server File Permissions
MySQL Installer can secure the server's data directory by updating the permissions of files and folders located at:
C:\ProgramData\MySQL\MySQL Server 8.0\Data
Do you want MySQL Installer to update the server file permissions for you?
Yes, grant full access to the user running the Windows Service (if applicable) and the administrators group only. Other users and groups will not have access.
Yes, but let me review and configure the level of access.
No, I will manage the permissions after the server configuration.
< Back
Next >
Cancel

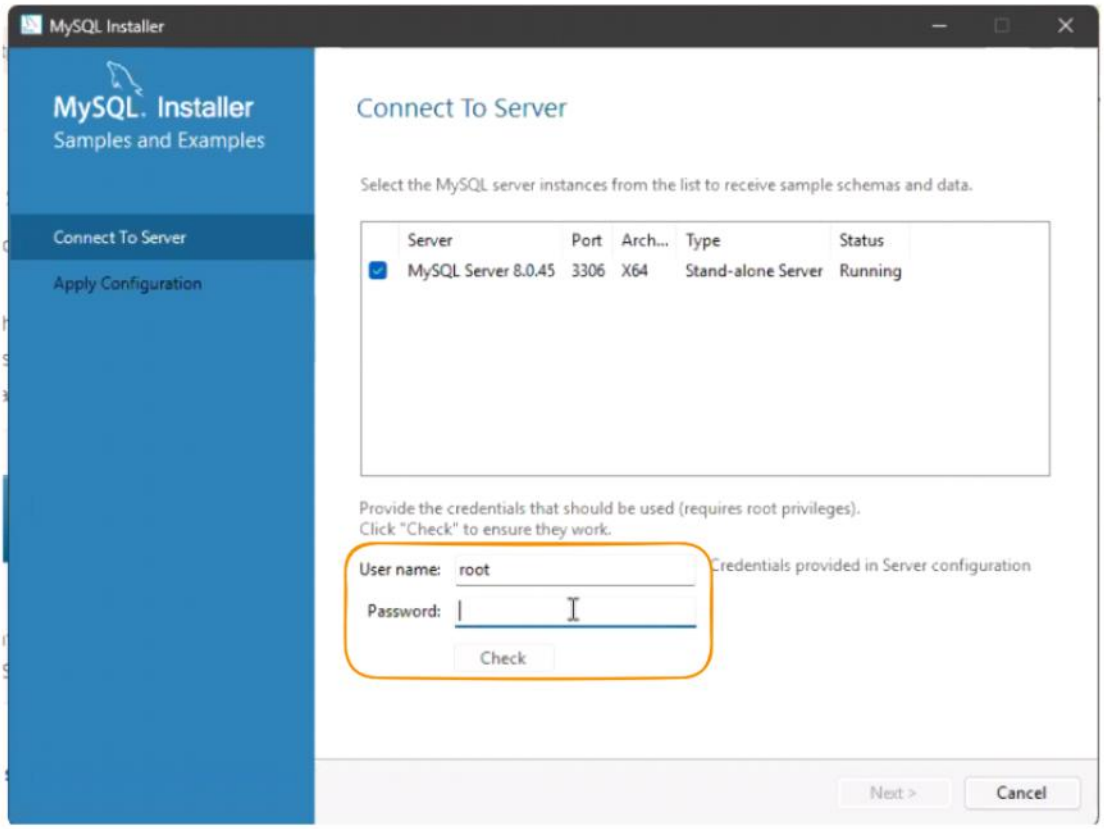

MySQL Installer
MySQL. Installer
MySQL Server 8.0.45
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
Click [Execute] to apply the changes
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
< Back
Execute
Cancel

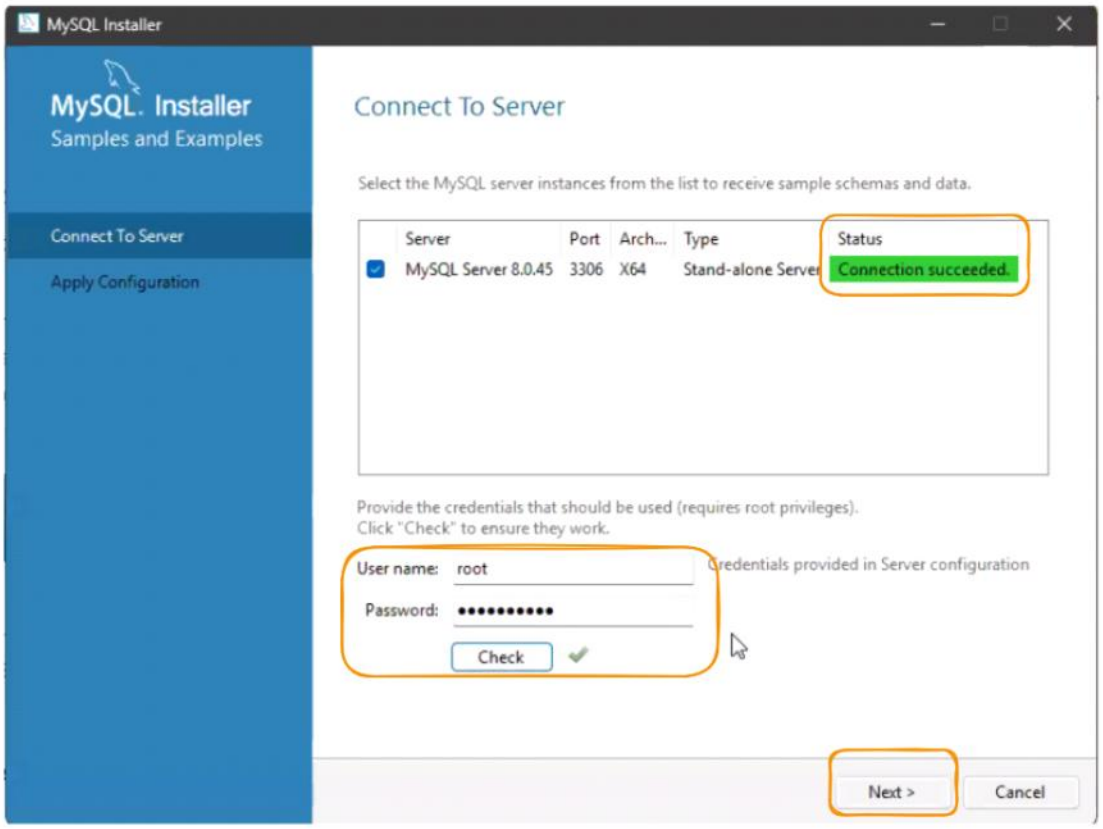

MySQL Installer
MySQL. Installer
MySQL Server 8.0.45
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
The configuration operation has completed.
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
The configuration for MySQL Server 8.0.45 was successful.
Click Finish to continue.
Finish

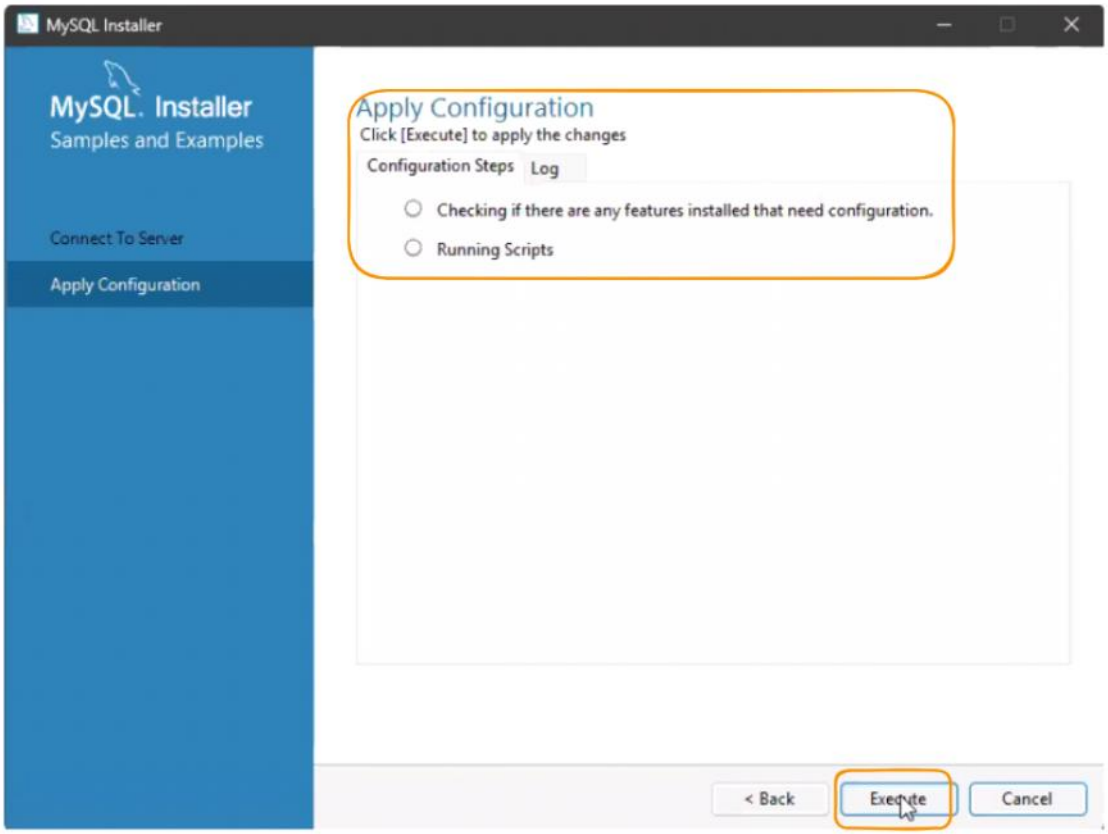

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.45
Configuration complete.
MySQL Router 8.0.45
Ready to configure
Samples and Examples 8.0.45
Ready to configure
Next >
Cancel

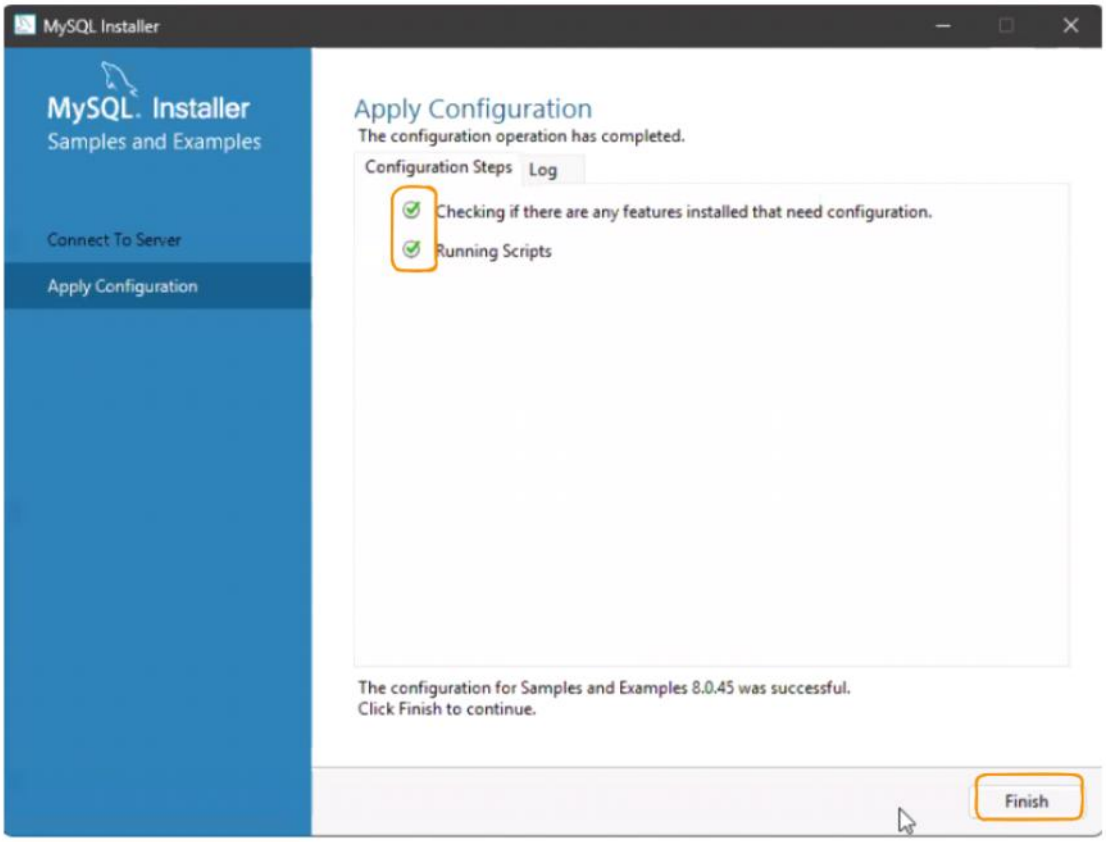

MySQL Installer
MySQL. Installer
MySQL Router 8.0.45
MySQL Router Configuration
MySQL Router Configuration
Bootstrap MySQL Router for use with InnoDB Cluster
This wizard can bootstrap MySQL Router to direct traffic between MySQL applications and InnoDB Cluster. Applications that connect to the router will be automatically directed to an available read/write or read-only member of the cluster.
The boostrapping process requires a connection to InnoDB Cluster. In order to register the MySQL Router for monitoring, use the current Read/Write instance of the cluster.
Hostname:
Port: 3310
Management User: root
Password:
Test Connection
MySQL Router requires specification of a base port (between 80 and 65532). The first port is used for classic read/write connections. The other ports are computed sequentially after the first port. If any port is indicated to be in use, please change the base port.
Classic MySQL protocol connections to InnoDB Cluster:
Read/Write: 6446
Read Only: 6447
X Protocol connections to InnoDB Cluster:
Read/Write: 6448
Read Only: 6449
Finish
Cancel

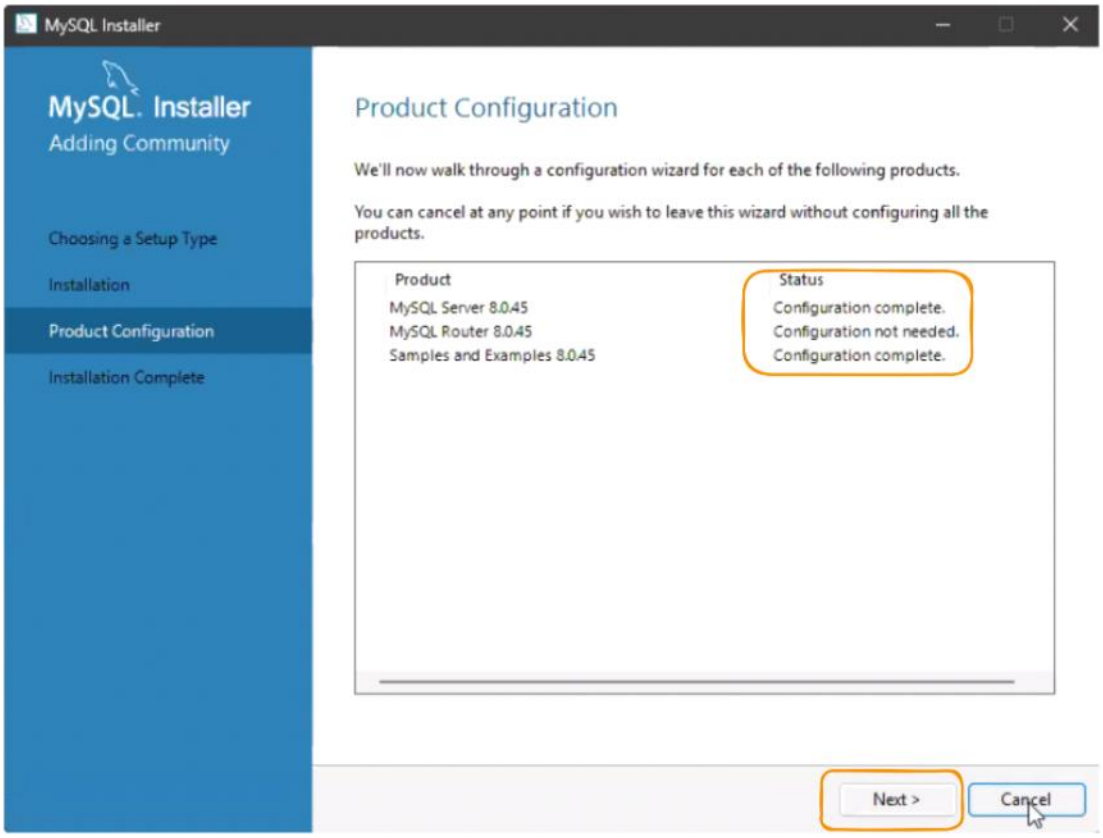

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.45
Configuration complete.
MySQL Router 8.0.45
Configuration not needed.
Samples and Examples 8.0.45
Ready to configure
Next >
Cancel

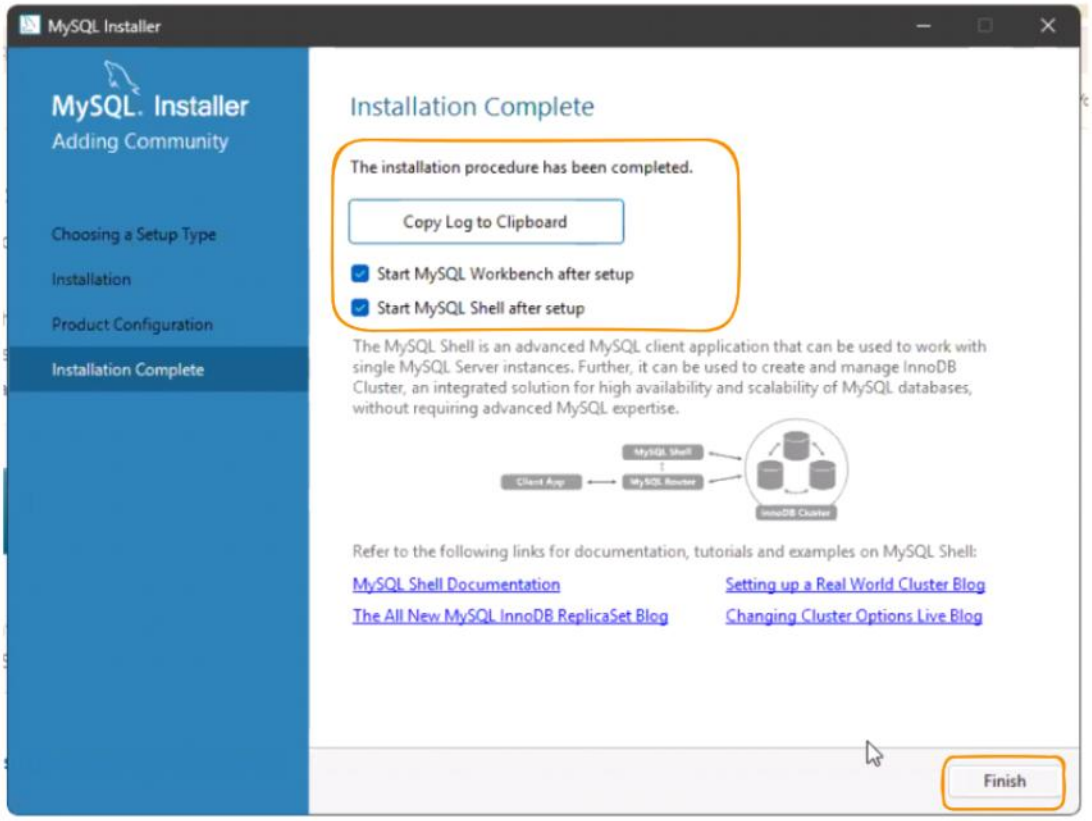

MySQL Installer
MySQL. Installer
Samples and Examples
Connect To Server
Apply Configuration
Connect To Server
Select the MySQL server instances from the list to receive sample schemas and data.
Server
Port
Arch...
Type
Status
MySQL Server 8.0.45
3306
X64
Stand-alone Server
Running
Provide the credentials that should be used (requires root privileges).
Click "Check" to ensure they work.
User name:
root
Password:
Check
Credentials provided in Server configuration
Next >
Cancel

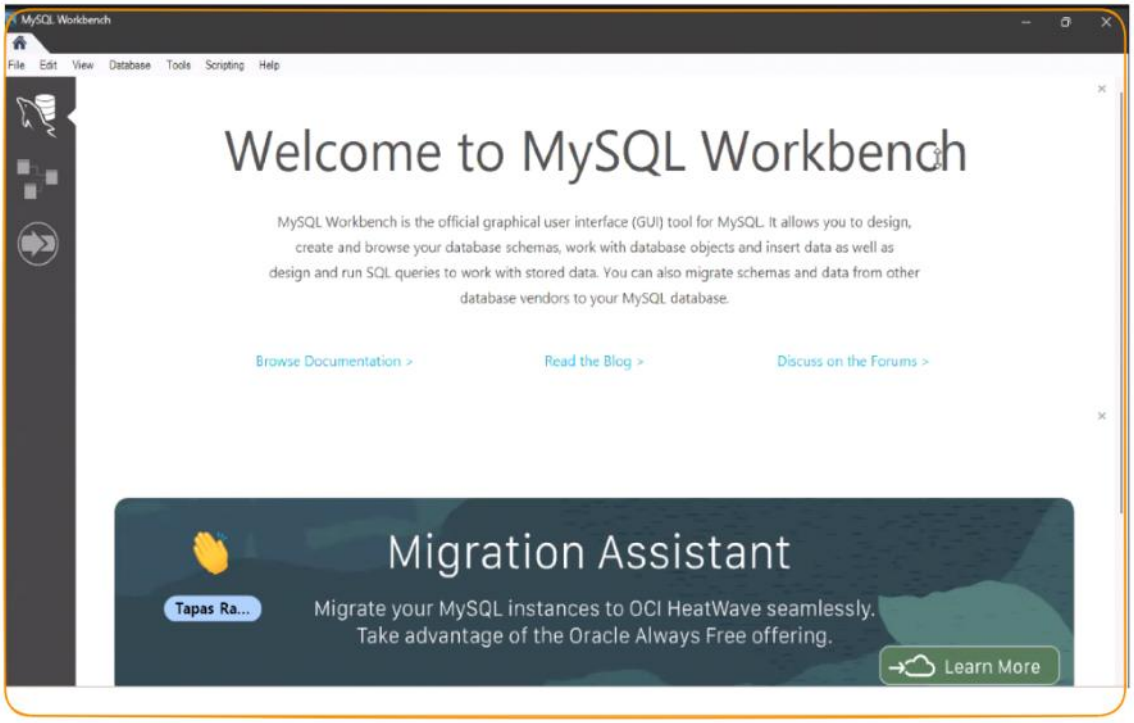

MySQL Installer
MySQL. Installer
Samples and Examples
Connect To Server
Apply Configuration
Connect To Server
Select the MySQL server instances from the list to receive sample schemas and data.
Server
Port
Arch...
Type
Status
MySQL Server 8.0.45
3306
X64
Stand-alone Server
Connection succeeded.
Provide the credentials that should be used (requires root privileges).
Click "Check" to ensure they work.
User name:
root
Password:
Check
Credentials provided in Server configuration
Next >
Cancel


MySQL Installer
MySQL. Installer
Samples and Examples
Connect To Server
Apply Configuration
Apply Configuration
Click [Execute] to apply the changes
Configuration Steps
Log
Checking if there are any features installed that need configuration.
Running Scripts
< Back
Execute
Cancel


MySQL Installer
MySQL. Installer
Samples and Examples
Connect To Server
Apply Configuration
Apply Configuration
The configuration operation has completed.
Configuration Steps
Log
Checking if there are any features installed that need configuration.
Running Scripts
The configuration for Samples and Examples 8.0.45 was successful.
Click Finish to continue.
Finish


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.45
Configuration complete.
MySQL Router 8.0.45
Configuration not needed.
Samples and Examples 8.0.45
Configuration complete.
Next >
Cancel


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Installation
Product Configuration
Installation Complete
Installation Complete
The installation procedure has been completed.
Copy Log to Clipboard
Start MySQL Workbench after setup
Start MySQL Shell after setup
The MySQL Shell is an advanced MySQL client application that can be used to work with single MySQL Server instances. Further, it can be used to create and manage InnoDB Cluster, an integrated solution for high availability and scalability of MySQL databases, without requiring advanced MySQL expertise.
Refer to the following links for documentation, tutorials and examples on MySQL Shell:
MySQL Shell Documentation
Setting up a Real World Cluster Blog
The All New MySQL InnoDB ReplicaSet Blog
Changing Cluster Options Live Blog
Finish


MySQL Workbench
File Edit View Database Tools Scripting Help
Welcome to MySQL Workbench
MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other database vendors to your MySQL database.
Browse Documentation >
Read the Blog >
Discuss on the Forums >
Migration Assistant
Tapas Ra...
Migrate your MySQL instances to OCI HeatWave seamlessly.
Take advantage of the Oracle Always Free offering.
Learn More